

International Cybersecurity and Digital Forensics Academy (ICDFA)

CIP-B101: Basic Computer Skills for Digital Forensics

Lab 1B: PC System Fundamentals, Disk Sectors and Partition Analysis

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Platform	Windows 11 Pro (DESKTOP-KSBSLAR) + Kali Linux VM

1. Introduction

The purpose of this lab is to develop a foundational understanding of PC system components and how they relate to digital forensic investigation. A junior digital forensic trainee must be able to identify hardware and software specifications, map where evidence can exist across system layers, understand how processes and thread's function, calculate disk and partition sizes from raw sector data, inspect partition tables using safe read-only commands, explain the PC boot process, and identify the file systems used by storage volumes.

This lab was carried out on a Windows 11 Pro system (DESKTOP-KSBSLAR) using built-in tools such as System Information (msinfo32), system info, PowerShell (Get-Disk, Get-Partition, Get-Volume, Get-Process), Disk Management (diskmgmt.msc), and DiskPart. A Kali Linux VM was also used for Linux-side partition inspection using lsblk, fdisk, parted, blkid, and df. All commands were executed in read-only mode in accordance with the forensic mindset: observe, record, calculate, verify, and document without altering the evidence source.

2. Tools Used

Tool / Command	Platform	Purpose
msinfo32 (System Information)	Windows 11	System specification and BIOS mode
systeminfo	Windows 11 CMD	Full system and network information
Get-Disk / Get-Partition / Get-Volume	Windows PowerShell	Disk and partition inspection
Get-Process	Windows PowerShell	Running process observation
DiskPart (list disk / list partition)	Windows CMD (Admin)	Partition table listing
Disk Management (diskmgmt.msc)	Windows 11	Visual disk and partition layout
lsblk / fdisk / parted	Kali Linux VM	Linux partition inspection
blkid / df / lsblk -f	Kali Linux VM	File system identification
Python 3 / Calculator	Both	Manual partition size calculations

Part A – System Identification and Evidence Layer Mapping

A1. System Specification (msinfo32 – Screenshots 1 & 2)

The following screenshots were captured from msinfo32 showing the key system identifiers:

Item	Value
OS Name	Microsoft Windows 11 Pro
Version	10.0.26200 Build 26200
Other OS Description	Not Available
OS Manufacturer	Microsoft Corporation
System Name	DESKTOP-KSBSLAR
System Manufacturer	Micro-Star International Co., Ltd.
System Model	GV62 7RE
System Type	x64-based PC
System SKU	16J9.3
Processor	Intel(R) Core(TM) i5-7300HQ CPU @ 2.50GHz, 2501 Mhz, 4 Core(s), 4 Logic...
BIOS Version/Date	American Megatrends Inc. E16J9IMS.318, 5/22/2017
SMBIOS Version	3.0
Embedded Controller Version	255.255
BIOS Mode	UEFI
BaseBoard Manufacturer	Micro-Star International Co., Ltd.
BaseBoard Product	MS-16J9
BaseBoard Version	REV:1.0
Platform Role	Mobile
Secure Boot State	On
PCR7 Configuration	Elevation Required to View
Windows Directory	C:\WINDOWS
System Directory	C:\WINDOWS\system32
Boot Device	\Device\HarddiskVolume4
Locale	United Kingdom
Hardware Abstraction Layer	Version = "10.0.26100.1"
Username	DESKTOP-KSBSLAR\user
Time Zone	GMT Summer Time
Installed Physical Memory (RAM)	16.0 GB
Total Physical Memory	15.9 GB
Available Physical Memory	8.08 GB
Total Virtual Memory	16.9 GB

Screenshot 1: msinfo32 – OS Name, Version, System Type, Processor, BIOS Mode, Memory, Secure Boot

Time Zone	GMT Summer Time
Installed Physical Memory (RAM)	16.0 GB
Total Physical Memory	15.9 GB
Available Physical Memory	8.08 GB
Total Virtual Memory	16.9 GB
Available Virtual Memory	7.00 GB
Page File Space	1.00 GB
Page File	C:\pagefile.sys
Kernel DMA Protection	Off
Virtualisation-based security	Running
Virtualisation-based security r...	
Virtualisation-based security a...	Base Virtualisation Support, Secure Boot, Mode Based Execution Control
Virtualisation-based security s...	
Virtualisation-based security s...	
App Control for Business policy	Enforced
App Control for Business user ...	Audit
Automatic Device Encryption ...	Elevation Required to View
A hypervisor has been detecte...	

Screenshot 2: msinfo32 continued – Memory, Virtual Memory, Page File, Security features

Item	Value
OS Name	Microsoft Windows 11 Pro
OS Version	10.0.26200 Build 26200

System Type	x64-based PC
Processor	Intel(R) Core(TM) i5-7300HQ CPU @ 2.50GHz, 2501 MHz, 4 Core(s)
System Model	MSI GV62 7RE
BIOS Mode	UEFI
Secure Boot State	On
Installed RAM	16.0 GB
System Manufacturer	Micro-Star International Co., Ltd.

A1. systeminfo Command Output (Screenshots 3 & 4)

```
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\user>systeminfo

Host Name:                DESKTOP-KSBSLAR
OS Name:                  Microsoft Windows 11 Pro
OS Version:               10.0.26200 N/A Build 26200
OS Manufacturer:         Microsoft Corporation
OS Configuration:        Standalone Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:         user
Registered Organization:   N/A
Product ID:               00330-52274-33924-AAOEM
Original Install Date:     5/27/2026, 2:19:03 PM
System Boot Time:          6/23/2026, 10:22:33 AM
System Manufacturer:       Micro-Star International Co., Ltd.
System Model:              GV62 7RE
System Type:               x64-based PC
Processor(s):              1 Processor(s) Installed.
                           [01]: Intel64 Family 6 Model 158 Stepping 9 GenuineIntel ~1500 Mhz
BIOS Version:              American Megatrends Inc. E16J9IMS.318, 5/22/2017
Windows Directory:         C:\WINDOWS
System Directory:          C:\WINDOWS\system32
Boot Device:               \Device\HarddiskVolume4
System Locale:              en-gb;English (United Kingdom)
Input Locale:              en-gb;English (United Kingdom)
Time Zone:                 (UTC+00:00) Dublin, Edinburgh, Lisbon, London
Total Physical Memory:      16,304 MB
Available Physical Memory:  7,925 MB
```

Screenshot 3: systeminfo – Host Name, OS details, Processor, Boot Device, Time Zone

```
Total Physical Memory:      16,304 MB
Available Physical Memory:  7,925 MB
Virtual Memory: Max Size:   17,328 MB
Virtual Memory: Available:  6,818 MB
Virtual Memory: In Use:      10,510 MB
Page File Location(s):      C:\pagefile.sys
Domain:                      WORKGROUP
Logon Server:                \\DESKTOP-KSBSLAR
Hotfix(s):                   4 Hotfix(s) Installed.
                           [01]: KB5087051
                           [02]: KB5054156
                           [03]: KB5094126
                           [04]: KB5094135
Network Card(s):             5 NIC(s) Installed.
                           [01]: Intel(R) Dual Band Wireless-AC 3168
                               Connection Name: WiFi 2
                               Status: Media disconnected
                           [02]: Qualcomm Atheros AR8171/8175 PCI-E Gigabit Ethernet Controller (NDIS 6.30)
                               Connection Name: Ethernet 2
                               Status: Media disconnected
                           [03]: Bluetooth Device (Personal Area Network)
                               Connection Name: Bluetooth Network Connection
                               Status: Media disconnected
                           [04]: VMware Virtual Ethernet Adapter for VMnet1
                               Connection Name: VMware Network Adapter VMnet1
                               DHCP Enabled: Yes
                               DHCP Server: 192.168.48.254
                               IP address(es)
                               [01]: 192.168.48.1
                               [02]: fe80::e32e:e870:c519:c358
                           [05]: VMware Virtual Ethernet Adapter for VMnet8
                               Connection Name: VMware Network Adapter VMnet8
                               DHCP Enabled: Yes
                               DHCP Server: 192.168.44.254
                               IP address(es)
                               [01]: 192.168.44.1
                               [02]: fe80::1190:cdb7:7f03:91e0
Virtualization-based security: Status: Running
Required Security Properties:
Available Security Properties:
```

Screenshot 4: systeminfo continued – Memory, Hotfixes, Network Cards

A2. Storage Information – Get-Disk, Get-Partition, Get-Volume (Screenshots 5, 6, 7)

```
Copyright (C) Microsoft Corporation. All rights reserved.
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows
PS C:\WINDOWS\system32> Get-Disk
```

Number	Friendly Name	Serial Number	HealthStatus	OperationalStatus	Total Size	Partition Style
0	HGST HTS54...	JD100A1V2VH3AM	Healthy	Online	931.51 GB	MBR
1	SAMSUNG MZ...	0025_386C_F100_1E84.	Healthy	Online	238.47 GB	GPT

Screenshot 5: Get-Disk – Two disks detected: Disk 0 (HGST HTS54, 931.51 GB, MBR) and Disk 1 (Samsung MZ, 238.47 GB, GPT)

```
PS C:\WINDOWS\system32> Get-Partition
```

DiskPath: \\?\scsi#disk&ven_hgst&prod_hts541010a9e680#4&2898b4be&0&020000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber	DriveLetter	Offset	Size	Type
1	D	1048576	50 MB	IFS
2	E	53477376	930.93 GB	IFS
3		999628472320	547 MB	Unknown

DiskPath: \\?\scsi#disk&ven_nvme&prod_samsung_mzvpv256#5&aofd18ff&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber	DriveLetter	Offset	Size	Type
1		1048576	200 MB	System
2		210763776	16 MB	Reserved
3	C	227540992	237.46 GB	Basic
4		255201378304	818 MB	Recovery

Screenshot 6: Get-Partition – Partition layout for both disks

```
PS C:\WINDOWS\system32> Get-Volume
```

DriveLetter	FriendlyName	FileSystemType	DriveType	HealthStatus	OperationalStatus	SizeRemaining	Size
C		NTFS	Fixed	Healthy	OK	151.68 GB	237.46 GB
D	System Reserved	NTFS	Fixed	Healthy	OK	19.64 MB	50 MB
E		NTFS	Fixed	Healthy	OK	818.54 GB	930.93 GB
		NTFS	Fixed	Healthy	OK	81.71 MB	547 MB
		NTFS	Fixed	Healthy	OK	114.85 MB	818 MB

Screenshot 7: Get-Volume – All volumes with file system type and sizes

A3. Evidence Layer Mapping Table

Layer / Source	Example Evidence	Where It May Be Found	Why It Matters
User layer	Opened files, search terms, typed usernames	Recent files list, browser history, shell/PowerShell history	Links user actions directly to system activity and timeline
Application layer	Office document metadata, browser cache, chat logs	Application folders, SQLite databases, AppData directories	Shows what software created or processed the evidence

OS layer	Windows Event Logs, installed programs, user profiles, registry entries	C:\Windows\System32\winevt\Logs, HKLM registry hive, C:\Users	Records system events, logon sessions, and account activity
Disk / Storage layer	Deleted files, file fragments, MFT records, partition metadata	NTFS MFT, unallocated space, file slack, volume shadow copies	Supports file recovery, timeline construction, and artifact analysis
Memory / Process layer	Running malware PIDs, network connections, injected code	RAM capture, process list (tasklist / Get-Process), pagefile.sys	Reveals live system behavior that disappears after shutdown
Network layer	IP connections, DNS queries, network traffic	Windows Firewall logs, Wireshark captures, router logs	Identifies communication with external hosts during an incident

Evidence may exist in more than one layer because a single user action typically triggers multiple system components simultaneously. For example, opening a browser downloads files to the disk layer, creates a process in memory, generates a Windows Event Log entry at the OS layer, and leaves behind application cache at the application layer. A forensic examiner must inspect all layers to build a complete and accurate picture of events.

Part B – Program, Process and Thread Investigation

B1. Running Processes – Get-Process (Screenshot 8)

```
PS C:\WINDOWS\system32> Get-Process | Select-Object -First 20 Name,Id,CPU,WorkingSet

Name                Id  CPU    WorkingSet
----
AggregatorHost      6792      14454784
ai                  1292 0.375   41009152
AppActions          16984 0.9375  45215744
ApplicationFrameHost 15736 0.265625 42926080
backgroundTaskHost   9068 0.34375  2514944
backgroundTaskHost  16424 0.671875 58875904
backgroundTaskHost  21264 0.65625  3256320
chrome              1868 13.0625 185110528
chrome              5544 3.640625 45641728
chrome              9488 0.109375 27336704
chrome             14020 0.0625   9478144
chrome             14136 0.765625 66306048
chrome             15332 10.9375  87097344
chrome             16520 0.203125 19017728
chrome             17892 26.84375 224411648
cmd                 9756 0.015625  8204288
CmWebAdmin          4812      17244160
CodeMeter            4904      51499008
CodeMeterCC         2520 7.03125 23859200
com.docker.backend  7052 3.125   63864832
```

Screenshot 8: Get-Process | Select-Object -First 20 Name,Id,CPU,WorkingSet

Process Name	PID	CPU (s)	Likely Purpose
chrome	1868	13.0625	Google Chrome browser – user is actively browsing the web
cmd	9756	0.015625	Windows Command Prompt – used to run systeminfo and DiskPart commands in this lab
com.docker.backend	7052	3.125	Docker Desktop backend service – managing container virtualisation

B2. Program, Process, and Thread – Explanation

Program:

A program is a static set of instructions stored on disk. It exists as an executable file such as chrome.exe or cmd.exe. On its own it does nothing it only becomes active when launched by the operating system.

Process:

A process is a running instance of a program. When chrome.exe is launched, the OS creates a process and assigns it a unique Process ID (PID). The process has its own dedicated memory space, handles, and resources. Multiple instances of Chrome each appear as separate processes with different PIDs in the screenshot above.

Thread:

A thread is the smallest unit of execution within a process. A single process can contain many threads running concurrently. For example, one Chrome thread may handle rendering while another handles network requests. Threads within the same process share memory space but have their own execution state.

Forensic Relevance:

Process information is critical during malware analysis and live forensic triage. Suspicious processes with unusual parent-child relationships, unexpected names, high CPU usage, or connections to external IP addresses can indicate compromise. PIDs allow correlation between process activity and event logs. Inspecting running processes before shutdown can reveal malware that lives only in memory and disappears when the system is powered off. The Working Set column shows how much RAM each process is using, which can identify memory-resident threats.

Part C – Disk Sector and Disk Size Calculation

C1. Slide Reproduction Calculation

Values from the course slide: sectors per track = 400, heads = 12, cylinders = 17,000, bytes per sector = 512

Item	Formula / Value	Answer
Total sectors	$400 \times 12 \times 17,000$	81,600,000 sectors
Total bytes	$81,600,000 \times 512$	41,779,200,000 bytes
Kilobytes	$41,779,200,000 \div 1024$	40,800,000 KB
Megabytes	$40,800,000 \div 1024$	39,843.75 MB
Gigabytes	$39,843.75 \div 1024$	38.91 GB

C2. Partition Size Calculations from This System

Disk 0 (HGST HTS54, MBR, 931.51 GB) – Partition 1 and Partition 2 are used for the Part G final assignment. The sector size for all disks is 512 bytes (confirmed by fdisk in the Kali VM which shows the same physical sector standard).

Disk 0 – Partition 1 (System Reserved, Drive D, 50 MB):

Reported size: 50 MB (DiskPart) / 50 MB (Disk Management)

Offset: 1,048,576 bytes = 1 MB from the start of the disk

Step 1: Convert size to bytes: $50 \text{ MB} \times 1,024 \times 1,024 = 52,428,800$ bytes

Step 2: Calculate sector count: $52,428,800 \div 512 = 102,400$ sectors

Step 3: Convert to GB: $52,428,800 \div 1,024 \div 1,024 \div 1,024 = 0.0488 \text{ GB} \approx 50 \text{ MB}$

Disk 0 – Partition 2 (Primary Data, Drive E, 930 GB):

Reported size: 930 GB (DiskPart) / 930.93 GB (Get-Partition)

Offset: 53,477,376 bytes = ≈ 51 MB from start

Step 1: Convert size to bytes (using 930 GB): $930 \times 1,024 \times 1,024 \times 1,024 = 998,579,896,320$ bytes

Step 2: Calculate sector count: $998,579,896,320 \div 512 = 1,950,351,360$ sectors

Step 3: Verify GB: $998,579,896,320 \div 1,024^3 = 930.00 \text{ GB}$

Kali Linux VM – /dev/sda1 (ext4, 80.1 GiB):

From fdisk: Start = 2,048, End = 167,968,749, Sectors = 167,966,702, Sector size = 512 bytes

Step 1: Total bytes = $167,966,702 \times 512 = 85,998,951,424$ bytes

Step 2: Convert to GB: $85,998,951,424 \div 1,024 \div 1,024 \div 1,024 = 80.09$ GB

df -h reports 79 GB usable – the difference is due to ext4 reserving approximately 5% of the partition for root use and filesystem metadata (journal, inode tables, superblock).

Part D – Safe Partition Table Inspection

D1. Windows – Disk Management (Screenshot 9)

The screenshot shows the Windows Disk Management console. At the top, a table lists all volumes. Below, two disks are shown: Disk 0 (931.51 GB, MBR) and Disk 1 (238.46 GB, GPT). Disk 0 has three partitions: System Reserved (D:), (E:), and a Recovery Partition. Disk 1 has four partitions: EFI System Partition, Microsoft Reserved, (C:), and a Recovery Partition.

Volume	Layout	Type	File System	Status	Capacity	Free Sp...	% Free
(C:)	Simple	Basic	NTFS (BitLo...	Healthy (B...	237.46 GB	151.67 GB	64 %
(E:)	Simple	Basic	NTFS	Healthy (P...	930.93 GB	818.54 GB	88 %
(Disk 0 partition 3)	Simple	Basic		Healthy (R...	547 MB	547 MB	100 %
(Disk 1 partition 1)	Simple	Basic		Healthy (E...	200 MB	200 MB	100 %
(Disk 1 partition 4)	Simple	Basic		Healthy (R...	818 MB	818 MB	100 %
System Reserved ...	Simple	Basic	NTFS	Healthy (A...	50 MB	20 MB	40 %

Disk	Partition	File System	Status	Capacity	Free Space	% Free
Disk 0 931.51 GB Online	System Reserved (D:)	50 MB NTFS	Healthy (Active, Primary Partition)	50 MB	20 MB	40 %
	(E:)	930.93 GB NTFS	Healthy (Primary Partition)	930.93 GB	818.54 GB	88 %
		547 MB	Healthy (Recovery Partition)	547 MB	547 MB	100 %
Disk 1 238.46 GB Online		200 MB	Healthy (EFI System Partition)	200 MB	200 MB	100 %
		16 MB	Microsoft Reserved	16 MB	16 MB	100 %
	(C:)	237.46 GB NTFS (BitLocker Encrypted)	Healthy (Boot, Page File, Crash Dump, Basic Data Partition)	237.46 GB	151.67 GB	64 %
		818 MB	Healthy (Recovery Partition)	818 MB	818 MB	100 %

Screenshot 9: Disk Management showing Disk 0 (931.51 GB, MBR) with three partitions and Disk 1 (238.46 GB, GPT) with four partitions

Disk 0 (HGST, MBR) layout:

- Partition 1 (D:): System Reserved, 50 MB NTFS, Healthy (Active, Primary Partition)
- Partition 2 (E:): 930.93 GB NTFS, Healthy (Primary Partition)
- Partition 3: Recovery Partition, 547 MB, Healthy

Disk 1 (Samsung NVMe, GPT) layout:

- Partition 1: 200 MB, EFI System Partition
- Partition 2: 16 MB, Microsoft Reserved
- Partition 3 (C:): 237.46 GB NTFS, BitLocker Encrypted, Boot/Page File/Crash Dump
- Partition 4: 818 MB, Recovery Partition

D1. DiskPart – List Disk and List Partition (Screenshot 10)

```
Microsoft DiskPart version 10.0.26100.1150
Copyright (C) Microsoft Corporation.
On computer: DESKTOP-KSBSLAR

DISKPART> list disk

   Disk ###  Status       Size       Free       Dyn  Gpt
   -----  -
   Disk 0    Online        931 GB     1024 KB
   Disk 1    Online        238 GB     1024 KB          *
```

```
DISKPART> select disk 0

Disk 0 is now the selected disk.

DISKPART> list partition

   Partition ###  Type              Size       Offset
   -----  -
   Partition 1    Primary           50 MB     1024 KB
   Partition 2    Primary          930 GB      51 MB
   Partition 3    Recovery          547 MB     930 GB

DISKPART> exit
```

Screenshot 10: DiskPart showing two disks (Disk 0 = 931 GB MBR, Disk 1 = 238 GB GPT*) and Disk 0 partition table (Partitions 1, 2, 3)

DiskPart confirms Disk 1 has the GPT flag (*) and Disk 0 uses MBR (no * in Gpt column). Disk 0 contains three partitions: Partition 1 Primary (50 MB), Partition 2 Primary (930 GB), and Partition 3 Recovery (547 MB).

D2. Linux/Kali – lsblk, fdisk, parted (Screenshots 11 & 12)

```
(kali㉿kali)-[~]
$ lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINT
NAME        SIZE TYPE FSTYPE MOUNTPOINT
sda          80.1G disk
└─sda1 80.1G part ext4 /
```

```
(kali㉿kali)-[~]
$ sudo fdisk -l
[sudo] password for kali:
Disk /dev/sda: 80.09 GiB, 86000000000 bytes, 167968750 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7ae167e4

Device       Boot Start      End  Sectors  Size Id Type
/dev/sda1    *      2048 167968749 167966702 80.1G 83 Linux
```

Screenshot 11: lsblk showing /dev/sda (80.1G disk) with sda1 (80.1G, ext4, mounted at /). fdisk -l confirms 167,968,750 sectors at 512 bytes, partition start=2048, end=167968749, sectors=167966702

```
(kali㉿kali)-[~]
$ sudo parted -l
Model: VMware, VMware Virtual S (scsi)
Disk /dev/sda: 86.0GB
Sector size (logical/physical): 512B/512B
Partition Table: msdos
Disk Flags:

Number  Start   End     Size    Type     File system  Flags
  1      1049kB  86.0GB  86.0GB  primary  ext4         boot
```

Screenshot 12: parted -l confirming Disk /dev/sda = 86.0 GB, sector size 512B/512B, partition table msdos, Partition 1 from 1049kB to 86.0GB (ext4, boot flag)

D2. Linux – blkid, df -h, lsblk -f (Screenshots 13 & 14)

```
(kali㉿kali)-[~]
$ blkid

(kali㉿kali)-[~]
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            879M   0    879M   0% /dev
tmpfs           197M   1.3M  196M   1% /run
/dev/sda1       79G   31G   44G   42% /
tmpfs           982M   4.0K  982M   1% /dev/shm
none            1.0M   0    1.0M   0% /run/credentials/systemd-journald.service
tmpfs           982M   28K   982M   1% /tmp
none            1.0M   0    1.0M   0% /run/credentials/getty@tty1.service
tmpfs           197M   112K  197M   1% /run/user/1000
none            1.0M   0    1.0M   0% /run/credentials/systemd-networkd.service

(kali㉿kali)-[~]
$
```

Screenshot 13: blkid (no UUID output visible; device not decorated) and df -h showing /dev/sda1 = 79G total, 31G used, 44G available (42% used), mounted at /

```
(kali㉿kali)-[~]
$ lsblk -f
blkid
df -Th

NAME      FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
sda
└─sda1 ext4   1.0   root 7182a487-ea9b-4cef-aaee-da764906e8f6  43.7G   39% /
Filesystem      Type      Size  Used Avail Use% Mounted on
udev            devtmpfs  879M   0    879M   0% /dev
tmpfs           tmpfs     197M   1.3M  196M   1% /run
/dev/sda1       ext4      79G   31G   44G   42% /
tmpfs           tmpfs     982M   4.0K  982M   1% /dev/shm
none            tmpfs     1.0M   0    1.0M   0% /run/credentials/systemd-journald.service
tmpfs           tmpfs     982M   44K   982M   1% /tmp
none            tmpfs     1.0M   0    1.0M   0% /run/credentials/getty@tty1.service
tmpfs           tmpfs     197M   112K  197M   1% /run/user/1000
none            tmpfs     1.0M   0    1.0M   0% /run/credentials/systemd-networkd.service

(kali㉿kali)-[~]
$
```

Screenshot 14: lsblk -f showing sda1 as ext4, version 1.0, label=root, UUID=7182a487-ea9b-4cef-aaee-da764906e8f6. df -Th confirms ext4 type, 79G size

Part E – MBR, GPT and Boot Process Explanation

E1. PC Boot Process – Numbered Sequence

The system under examination uses UEFI mode (confirmed by msinfo32 BIOS Mode = UEFI and the presence of an EFI System Partition on Disk 1).

#	Stage	What Happens	Forensic Relevance
1	Power-on and POST	Hardware is powered and the CPU runs the Power-On Self Test. Memory, CPU, and storage devices are checked.	Hardware failure or physical tampering may prevent evidence access or corrupt data.
2	UEFI Firmware Initialisation	UEFI firmware loads from the motherboard chip. Boot device order from CMOS/NVRAM settings is applied.	UEFI settings can reveal boot order manipulation or Secure Boot bypass attempts.
3	Secure Boot Check	UEFI verifies the boot loader signature against the Secure Boot database. Unsigned code is rejected.	Secure Boot On (confirmed) means unauthorised bootloaders cannot run. Evidence of Secure Boot bypass is significant.
4	EFI Boot Entry / GPT	UEFI reads the EFI System Partition (200 MB on Disk 1, Partition 1) and locates the EFI boot loader (e.g., bootmgfw.efi).	EFI partition contents can contain boot configuration artifacts and evidence of dual-boot or hidden OS configurations.
5	Boot Loader / Boot Manager	Windows Boot Manager (bootmgr) loads and presents OS selection if multiple OS entries exist. Pre-kernel environment starts.	Boot configuration data (BCD store) can show installed operating systems and boot modifications.
6	Kernel Loading	The Windows kernel (ntoskrnl.exe) and HAL load. Device drivers initialise. Services start.	Kernel crash dumps and driver load order can reveal rootkits or driver-level malware.
7	OS Control and User Session	Windows Session Manager (smss.exe), WinLogon, and the user desktop load. The OS is fully operational.	Event logs, registry hives, and user activity begin to accumulate from this point.

This system uses UEFI boot mode. The boot device is \Device\HarddiskVolume4 (confirmed in systeminfo), which corresponds to the C: partition on Disk 1 (NVMe Samsung, GPT). Secure Boot is ON.

E2. MBR vs GPT Comparison

Feature	MBR	GPT
Age / Design	Older standard from the 1980s; used with BIOS/Legacy boot.	Modern standard used with UEFI systems.
Disk Size Limit	Commonly limited to approximately 2 TB.	Supports very large disks; theoretical limit approximately 9.4 ZB.
Partition Count	Maximum 4 primary partitions (or 3 primary + 1 extended).	Up to 128 partitions on Windows; no extended partition needed.
Partition Table Location	Stored in sector 0 (first 512 bytes), combined with boot code.	Primary GPT header at sector 1; backup GPT header at last sector.
Redundancy	No backup; damage to sector 0 can destroy partition information.	Backup GPT header aids recovery and cross-validation.
Forensic Issue	Critical evidence at sector 0; corruption can prevent partition interpretation.	Backup structures may assist recovery and GPT header validation.
This System	Disk 0 (HGST HDD, 931 GB) uses MBR.	Disk 1 (Samsung NVMe SSD, 238 GB, boot disk) uses GPT.

Part F – File System Identification and Interpretation

File Systems Identified on This System

Drive/Device	Volume Name	File System	Size	Observation
C:	(none)	NTFS (BitLocker)	237.46 GB	Boot partition, encrypted with BitLocker. NTFS MFT and journals present.
D:	System Reserved	NTFS	50 MB	Active primary partition on Disk 0. Contains boot files for legacy recovery.
E:	(none)	NTFS	930.93 GB	Large primary data partition on Disk 0 (HGST HDD).
/dev/sda1	root	ext4	80.1 GiB	Kali Linux VM root partition. UUID: 7182a487-ea9b-4cef-aaee-da764906e8f6.

Why File Systems Matter in Digital Forensics

File System	Common OS / Usage	Key Feature	Forensic Importance
NTFS	Modern Windows	MFT, journals, timestamps, permissions, alternate data streams	Rich metadata for timeline analysis; ADS can hide data; journal aids deleted file recovery.
FAT/FAT32	Removable media, older systems	Simple directory table, no journaling	Easy to recover deleted files; widely used on USB drives recovered in investigations.
exFAT	Large USB/SD cards	Extended FAT for large files	Commonly found on camera memory cards; simpler metadata than NTFS.
ext4	Linux systems (Kali, Ubuntu)	Inode-based, journaling, extents	Inode timestamps and journal support forensic timeline reconstruction on Linux systems.
HFS+	Older macOS	Catalog B-tree, resource forks	Important in Mac investigations; metadata and timestamps differ from NTFS.

All Windows volumes on this system use NTFS. NTFS is forensically rich because it maintains a Master File Table (MFT) with entries for every file, including \$STANDARD_INFORMATION and \$FILE_NAME timestamps, file size, permissions, and alternate data streams. The NTFS journal (\$LogFile and \$UsnJrnl) records changes and can help recover deleted or modified file information. The BitLocker encryption on the C: drive means the contents are not readable without the decryption key a significant consideration in a forensic investigation.

The Kali Linux partition uses ext4, which is an inode-based journaling file system. Each file has an inode storing access, modification, and change timestamps, as well as permissions and ownership data. The journal (located in the /dev/sda1 partition) records metadata changes and can assist in recovering partially deleted data and understanding recent filesystem activity.

Part G – Final Practical Assignment: Partition Size Verification

The partition table used for this assignment is from Disk 0 (HGST HTS54, MBR, 931 GB) as inspected via DiskPart (Screenshot 10) and Get-Partition (Screenshot 6). Sector size = 512 bytes.

Task	Required Evidence	Answer
Identify Partition 1	Screenshot of partition table and command output	Disk 0, Partition 1: Type = Primary, Drive Letter = D, Size = 50 MB (DiskPart Screenshot 10 / Get-Partition Screenshot 6)
Record Partition 1 sector count	Start sector, end sector or sector count	Get-Partition offset = 1,048,576 bytes. Size = 50 MB = 52,428,800 bytes. Sector count = $52,428,800 \div 512 = 102,400$ sectors.
Calculate Partition 1 size	Formula and result in GB	$102,400 \times 512 = 52,428,800$ bytes $\div 1,024 \div 1,024 \div 1,024 = 0.0488$ GB ≈ 50 MB
Identify Partition 2	Screenshot of partition table and command output	Disk 0, Partition 2: Type = Primary, Drive Letter = E, Size = 930 GB (DiskPart) / 930.93 GB (Get-Partition). Offset = 53,477,376 bytes.
Record Partition 2 sector count	Start sector, end sector or sector count	Size = 930 GB = 998,579,896,320 bytes. Sector

		count = 998,579,896,320 ÷ 512 = 1,950,351,360 sectors.
Calculate Partition 2 size	Formula and result in GB	1,950,351,360 × 512 = 998,579,896,320 bytes ÷ 1,024 ÷ 1,024 ÷ 1,024 = 930.00 GB
Compare with OS-reported size	Disk Management, Get-Volume, lsblk, or df output	Get-Volume reports E: = 930.93 GB. Disk Management shows 930.93 GB. DiskPart shows 930 GB (rounded).
Explain difference if any	Short explanation of rounding, reserved space, or GiB/GB differences	DiskPart rounds to the nearest GB (930). Get-Partition and Disk Management report 930.93 GB because they use more precise byte counts. The small difference (~0.93 GB) is due to rounding at the GB boundary and partition alignment overhead. Windows reports sizes in decimal GB (1 GB = 1,000,000,000 bytes) in some contexts and binary GiB (1 GiB = 1,073,741,824 bytes) in others, which contributes to minor discrepancies.

Step-by-Step Calculation Summary

Partition 1 (D: System Reserved, 50 MB):

Size in MB: 50 MB

Step 1: $50 \times 1,024 \times 1,024 = 52,428,800$ bytes

Step 2: $52,428,800 \div 512 = 102,400$ sectors

Step 3: $52,428,800 \div 1,073,741,824 = 0.0488$ GB (approximately 50 MB)

Confirmed by: Get-Partition (50 MB), Disk Management (50 MB), Get-Volume (50 MB, NTFS, Drive D:)

Partition 2 (E: Primary Data, 930 GB / 930.93 GB):

Size from DiskPart: 930 GB

Step 1: $930 \times 1,024 \times 1,024 \times 1,024 = 998,579,896,320$ bytes

Step 2: $998,579,896,320 \div 512 = 1,950,351,360$ sectors

Step 3: $998,579,896,320 \div 1,073,741,824 = 930.00$ GB

Confirmed by: Get-Volume (930.93 GB NTFS, Drive E:), Disk Management (930.93 GB NTFS, Primary Partition)

Note: The 0.93 GB difference between DiskPart (930 GB) and Get-Partition (930.93 GB) is due to DiskPart rounding to the nearest GB integer while Get-Partition reports the precise byte-level partition size.

Screenshot Checklist

No.	Screenshot Required	Included?
1	System Information (msinfo32) output	Yes
2	system info command output	Yes
3	Get-Disk – disk drive model and size output	Yes
4	Get-Volume – logical volume/file system output	Yes
5	Get-Process – running process/PID output	Yes
6	Disk Management visual layout	Yes
7	DiskPart list disk / list partition output	Yes
8	Linux lsblk, fdisk -l output (Partition 1 evidence)	Yes
9	Linux parted -l output	Yes
10	blkid, df -h, lsblk -f – file system identification	Yes

Conclusion

This lab successfully documented the PC system fundamentals of the DESKTOP-KSBSLAR machine running Windows 11 Pro and its companion Kali Linux virtual machine. The key findings and lessons are summarised below.

The system runs a 64-bit Intel Core i5-7300HQ processor with 16 GB of RAM. It uses UEFI firmware with Secure Boot enabled, which is confirmed both in msinfo32 (BIOS Mode = UEFI) and by the presence of a 200 MB EFI System Partition on Disk 1. The operating system boots from the GPT-partitioned Samsung NVMe SSD (Disk 1), while the larger HGST HDD (Disk 0, MBR) is used for secondary data storage.

The disk and partition calculations demonstrated that the formula “sectors $\times$ bytes per sector = total bytes” is reliable for estimating partition sizes. Minor discrepancies between DiskPart, Disk Management, and Get-Partition values arise from integer rounding and the difference between decimal GB and binary GiB unit systems an important distinction that must be understood in forensic reporting.

The process investigation showed that modern systems run many concurrent processes, including multiple Chrome instances, Docker services, and system services. Understanding

PIDs and process relationships is critical during live forensic triage, particularly when investigating memory-resident malware.

All Windows volumes use NTFS, which provides rich metadata for forensic analysis including MFT entries, NTFS journals, timestamps, and alternate data streams. The BitLocker encryption on the C: drive highlights the importance of acquiring decryption keys before disk analysis. The Kali VM partition uses ext4, an inode-based journaling system that similarly supports forensic timeline reconstruction.

The most important lesson from this lab is that a forensic examiner must be able to independently verify storage device information from first principles using sector counts and the conversion formula not simply accept the sizes reported by the operating system. All observations in this lab were made using read-only commands in compliance with the forensic mindset.